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# Experiment 6: Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

## 1. Aim

To explore the cloud service models IaaS, PaaS and SaaS by using the free tier of a public cloud platform such as AWS, Microsoft Azure or Google Cloud, and to compare the level of control and responsibility at each model.

## 2. Introduction

Cloud computing delivers computing resources over the internet on demand, usually on a pay-as-you-go basis. According to the widely used NIST definition, cloud services have five essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity and measured service. The three main service models differ by how much of the technology stack is managed by the provider and how much is managed by the customer.

## 3. Comparison of Service Models

| Layer | IaaS | PaaS | SaaS |
|---|---|---|---|
| Application and data | Customer | Customer | Provider |
| Runtime and middleware | Customer | Provider | Provider |
| Operating system | Customer | Provider | Provider |
| Virtualisation, servers, storage, network | Provider | Provider | Provider |
| Example services | Virtual machines, object storage | App hosting, managed databases | Email, office suites, CRM |
| Typical user | System administrators, DevOps | Developers | End users and businesses |

## 4. Hands-on Exploration

IaaS: a free-tier virtual machine was created with a small Linux image, connected through SSH, and a basic web server was installed. This exercise showed that the customer is responsible for patching the operating system, configuring the firewall rules and monitoring the server.

PaaS: a sample web application was deployed to a managed application hosting service. The deployment needed only the application code and a configuration file. Scaling was a setting rather than a server task, and the operating system and runtime updates were handled by the platform.

SaaS: a web-based email or document application was used. The only customer tasks were creating an account, configuring users and organising data. No infrastructure work was needed at all.

## 5. Observations

- The IaaS virtual machine required the largest number of configuration steps, including key pair creation, security group rules and package installation.

- The PaaS deployment took fewer steps, but the application had to follow the runtime and file structure required by the platform.
- The SaaS application was ready to use in minutes, and its features could not be changed beyond the settings it offered.
- Free-tier limits such as compute hours, storage and data transfer were read carefully, and a billing alert was set to avoid unexpected charges.
- Resources created for the exercise were stopped and deleted after the study to avoid ongoing charges.

## 6. Analysis

Moving from IaaS to PaaS to SaaS transfers responsibility from the customer to the provider. The trade-off is control against convenience. IaaS gives the greatest freedom in choosing software and configuration, but it demands skills in operations and security. PaaS removes much of the operational burden and speeds up development, but it restricts the runtime and deployment options. SaaS removes almost all technical work, but the customer has the least ability to customise the product and depends fully on the provider for its continuity.

A cost perspective is also important. IaaS and PaaS charge for resources used, so idle capacity is paid for unless it is turned off. SaaS usually charges per user per month, which is predictable but can become expensive as the number of users grows.

## 7. Limitations

The exercise used only a small free-tier workload, so performance, reliability and cost at production scale could not be assessed. Each provider uses different names and interfaces for similar services, and the table above represents common categories rather than exact product mappings.

## 8. Result and Conclusion

The study shows that IaaS, PaaS and SaaS represent a spectrum of responsibility. An organisation chooses the model based on its need for control, its technical skills, its cost structure and how quickly it needs to deliver. Hands-on work on a free-tier platform makes these differences concrete and shows why many organisations combine models in one architecture.